Syllabus

Semester	Course Code	Name of the course	L	T	P	Credits
V	**CS5M004**	**Ethics, Bias and Safety in LLM**	2	0	0	**2**

Course Outcomes:

Sr. No	Course Outcome number	CO statement
1	CO1	Students will be able to Analyze ethical challenges in LLM development and deployment.
2	CO2	Students will be able to Detect and mitigate bias in training datasets and model outputs.
3	CO3	Students will be able to Assess and improve the safety of LLM-based applications.
4	CO4	Students will be able to Understand legal and regulatory frameworks surrounding AI usage.
5	CO5	Students will be able to Evaluate and propose solutions to ethical controversies in LLMs using real-world case studies.

Course Contents		Hours
Unit I	**Ethical Foundations in LLMs:** Introduction to AI ethics, Responsible AI principles, Ethics in NLP and LLMs: Accountability, transparency, and fairness, Ethical decision-making models, Introduction to fairness in machine learning.	05

Introduction to AI Ethics

AI Ethics:

AI Ethics is the field that studies how to develop and use artificial intelligence in a way that is fair, accountable, transparent, and respects human values.

In short, ethics is a subfield of philosophy that provides us with a framework for deciding what is moral and what is not. While this may sound like an abstract concept, it takes on a specific meaning in the realm of technology—AI ethics is the application of moral frameworks to the development and use of artificial intelligence.

Imagine a biased loan approval algorithm or an opaque facial recognition system. These are examples of how AI, without ethical considerations, can lead to unfair outcomes. AI ethics tackles these concerns head-on by establishing a set of guiding principles for responsible AI development.

These principles aim to promote fairness, accountability, transparency, and privacy throughout the AI lifecycle. Data scientists can ensure this powerful technology benefits everyone fairly and responsibly by understanding and applying AI ethics.

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Short-term Ethical Risks of AI

Bias

To ensure fair and unbiased AI models, the training data needs to be representative of the entire population the model will be used on, with particular attention to including minority groups. If the training data fails to reflect marginalized communities accurately, AI systems can produce biased outputs, further widening disparities in their results.

Security & data privacy risks

Biased input training data is just one of the many data challenges that AI systems face. Another critical concern arises when the training data includes sensitive information such as PI, health records, or financial details. A breach could expose this sensitive data, potentially leading to identity theft, financial fraud, and other malicious uses

Misinformation & disinformation

While long standing data issues such as misrepresentation and privacy persist, the emergence of generative AI applications has introduced new concerns, particularly regarding misinformation and disinformation.

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Long-term risks of AI

Job displacement

One of AI's core benefits is bringing efficiency by improving processes through automation. However, the harsh reality of such automation leads to job loss, particularly those involving low-skilled, repetitive, or routine work, such as customer service and administrative support.

Erosion of privacy & human autonomy

AI-powered facial and audio systems can analyze incoming feeds from surveillance devices to identify and track individuals in real time. However, such mass monitoring is often implied without consent, thereby raising concerns over privacy erosion.

AI misalignment & existential risk

As autonomous AI systems rapidly evolve, a critical debate is consuming industry professionals, AI veterans, and the public: are we hurtling towards "superi ntelligence?"
This hypothetical scenario envisions AI surpassing human cognitive abilities and potentially posing a threat. The core concern lies in the potential misalignment between AI objectives and human values.

Contd..Examples

An alarming incident of societal bias came to light when AI generated a code **predicting seniority based on race and gender**, putting minority groups at a disadvantage. how ChatGPT classified black females as junior, while black males are classified as mid-level. Furthermore, the model assumed white females to be mid-level (notice the upgrade from black females) and white males to be senior.

Let's consider another case. An AI model predicted that **black patients are less likely to need extra care**, declaring them healthier than white patients. Such racial bias deprived more than half of the black patients of necessary care.

predictive justice. Here, AI models have been used to assess a defendant's risk of committing future crimes. However, these models have raised concerns about bias, with some instances predicting a **higher likelihood of recidivism for Black defendants** compared to white defendants.

Contd..AI Basic Terms

AI: The ability of a machine to perform cognitive functions we associate with human minds, such as perceiving, reasoning, learning, and problem solving. There are two main types of AI systems, and some systems are a combination of both:

Rule-based AI, also called **expert AI**, behaves according to a set of fully defined rules created by human experts—as an example, many e-commerce platforms use rule-based AI to provide product recommendations

Learning-based AI solves problems and adapts its functionality on its own, based on its initial human-designed configuration and training dataset—generative AI tools are examples of learning-based AI

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AI ethics: A set of values, principles, and techniques that employ widely accepted standards of right and wrong to guide moral conduct in the development, deployment, use, and sale of AI technologies.

AI model: A mathematical framework created by people and trained on data that enables AI systems to perform certain tasks by identifying patterns, making decisions, and predicting outcomes. Common uses include image recognition and language translation, among many others.

AI system: A complex structure of algorithms and models designed to mimic human reasoning and perform tasks autonomously.

Agency: The capacity of individuals to act independently and to make free choices.

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Bias: An inclination or prejudice for or against a person or group, especially in a way considered to be unfair. Biases in training data—such as the under- or over-representation of data pertaining to a certain group—can cause AI to act in biased ways.

Explainability: The ability to answer the question, “What did the machine do to reach its output?” Explainability refers to the technological context of the AI system, such as its mechanics, rules and algorithms, and training data.

Fairness: Impartial and just treatment or behavior without unjust favoritism or discrimination.

Human-in-the-loop: The ability of human beings to intervene in every decision cycle of an AI system.

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Upholding data privacy and protection: AI systems must meet the most stringent data privacy and protection standards, using robust cyber security methods to avoid data breaches and unauthorized access

Promoting inclusivity and diversity: AI technologies need to reflect and respect the vast range of human identities and experiences

Society and economies: AI should help drive societal advancement and economic prosperity for all people, without fostering inequality or unfair practices

Enhancing digital skills and literacy: AI technologies should strive to be accessible and understandable to everyone, regardless of a person's digital skills

AI Principle

These principles commonly address:

Human wellbeing and dignity: AI systems should always prioritize and ensure the wellbeing, safety, and dignity of individuals, neither replacing humans nor compromising human welfare

The health of businesses: AI business technologies should accelerate processes, maximize efficiency, and promote growth **Human oversight:** AI needs human monitoring at every stage of development and use—sometimes called "a human in the loop"—to ensure that ultimate ethical responsibility rests with a human being

Addressing bias and discrimination: Design processes should prioritize fairness, equality, and representation to mitigate bias and discrimination

Transparency and explainability: How AI models make specific decisions and produce specific results should be transparent and explainable in clear language

Responsible AI -Principles

following broad principles for responsible management of AI:

1. Principle of Safety and Reliability
2. Principle of Equality
3. Principle of Inclusivity and Non-discrimination
4. Principle of Privacy and security
5. Principle of Transparency
6. Principle of Accountability
7. Principle of protection and reinforcement of positive human values

Responsible AI Principle

Responsible artificial intelligence (AI) is a set of principles that help guide the design, development, deployment and use of AI—building trust in AI solutions that have the potential to empower organizations and their stakeholders.

Responsible AI involves the consideration of a broader societal impact of AI systems and the measures required to align these technologies with stakeholder values, legal standards and ethical principles.

Responsible AI aims to embed such ethical principles into AI applications and workflows to mitigate risks and negative outcomes associated with the use of AI, while maximizing positive outcomes.

Explainability

Machine learning models such as deep neural networks are achieving impressive accuracy on various tasks. But explainability and interpretability are ever more essential for the development of trustworthy AI.

A. Prediction accuracy

Accuracy is a key component of how successful the use of AI is in everyday operation. By running simulations and comparing AI output to the results in the training data set, the prediction accuracy can be determined.

B. Traceability

Traceability is a property of AI that signifies whether it allows users to track its predictions and processes.

C. Decision understanding

This is the human factor. Practitioners need to be able to understand how and why AI derives conclusions.

Responsible AI Principle-cnt..

Fairness

Machine learning models are increasingly used to inform high stakes decision-making that relates to people. Although machine learning, by its very nature, is a form of statistical discrimination, the discrimination becomes objectionable when it places privileged groups at systematic advantage and certain unprivileged groups at systematic disadvantage, potentially causing varied harms.

Diverse and representative data
Bias-aware algorithms
Bias mitigation techniques
Diverse development teams
Ethical AI review boards

Responsible AI Principle-cnt..

Robustness

Robust AI effectively handles exceptional conditions, such as abnormalities in input or malicious attacks, without causing unintentional harm. It is also built to withstand intentional and unintentional interference by protecting against exposed vulnerabilities. Our increased reliance on these models and the value they represent as an accumulation of confidential and proprietary knowledge, are at increasing risk for attack.

Transparency

Users must be able to see how the service works, evaluate its functionality, and comprehend its strengths and limitations. Increased transparency provides information for AI consumers to better understand how the AI model or service was created.

Responsible AI Principle-cnt..

Privacy

Many regulatory frameworks, including GDPR, mandate that organizations abide by certain privacy principles when processing personal information. A malicious third party with access to a trained ML model, even without access to the training data itself, can still reveal sensitive personal information about the people whose data was used to train the model.

Responsible AI Principle-cnt..

Steps to Implement responsible AI practices

- Define responsible AI principles
- Educate and raise awareness
- Integrate ethics across the AI development lifecycle
- Protect user privacy
- Facilitate human oversight
- Encourage external collaboration

Ethics in NLP & LLM: Transparency

Transparency, accountability, and fairness-

These are crucial aspects of Natural Language Processing (NLP) and Large Language Models (LLMs) to ensure responsible and ethical use. These principles address potential biases, promote responsible development, and build trust in these powerful technologies.

Transparency in NLP and LLMs refers to the ability to understand how these models work, their decision-making processes, and the data they are trained on.

Algorithmic Transparency: Understanding the inner workings of the algorithms and models.

Data Transparency: Comprehending how data is collected, stored, and used.

Process Transparency: Understanding the development and deployment processes, including testing and validation.

Ethics in NLP & LLM: Fairness

Ethical decision making model

Strategies for Ethical Decision Making: Eight Step Model

Step One

Determine if Matter is and Ethical One

Locate the general ethical principles that apply to the situation to determine if the matter is ethical.

Ethical decision making model

Strategies for Ethical Decision Making: Eight Step Model

Step Two

Consult the guidelines already available that might apply to a specific identification and possible mechanism for resolution

Ethical decision making model

Strategies for Ethical Decision Making: Eight Step Model

Step Three

Consider, as best as possible, all of the sources that might influence the kind of decision you will make

- Evaluate prejudices, personal needs and attitudes

Ethical decision making model

Strategies for Ethical Decision Making: Eight Step Model

Step Four

Locate a trusted colleague with whom you can consult

- Evaluate consultants abilities and expertise
- Be aware of liability issues
- Consultation should be based on objectivity

Strategies for Ethical Decision Making: Eight Step Model

Step Five

Generate alternative decisions

Brain storm all options

Ethical decision making model

Strategies for Ethical Decision Making: Eight Step Model

Step Six

Enumerate consequences of each decision include economic, psychological, and social costs; short-term, ongoing, and long term effects; the time and effort necessary to effect each decision, including any resource limitations; any other risks, including the violation of individual rights; and any benefits.

Ethical decision making model

Strategies for Ethical Decision Making: Eight Step Model

Step Seven

Make the decision
Review decision with affected parties with regard to outcome effects

Ethical decision making model

Strategies for Ethical Decision Making: Eight Step Model

Step Eight

Implement the decision
Should do versus will do dichotomy
"Survey data have also revealed that clinical psychologists were most often willing to implement decisions that were less direct, less restrictive, and less consistent with APA guidelines, often acting from expedience and opportunism"

Fairness in Machine Learning

What is machine learning fairness?

Machine learning fairness is the process of correcting and eliminating algorithmic bias (of race and ethnicity, gender, sexual orientation, disability, and class) from machine learning models. Machine learning is a branch of artificial intelligence (AI) that stems from the idea that computers can learn from data collected to identify patterns and make decisions that mimic those of humans, with minimal human intervention.

Fairness in Machine Learning

Why is it important to address fairness in machine learning?

Unintentional discrimination in machine learning algorithms is a primary reason why it's important to address fairness and AI ethics.

Machine learning is becoming enmeshed in the systems and applications we use to help us buy furniture, find jobs, recruit new hires, apply for universities, listen to music, get loans, find news, search on Google, target ads, and so much more. It enhances our ability to streamline information and provide recommendations, but it has serious consequences if it receives training on the wrong information and fails to promote fair and equal practices.

To remove these potential biases, data scientists and machine learning experts must look out for them in algorithmic models and correct them. Because machine learning by definition learns by example, it can also "learn" to avoid bias as long as it receives the right data. Used in industries as varied as the criminal justice system to corporate human resources to credit lending, it's important that machine learning adopts fair and ethical processes.

Fairness in Machine Learning

3 ways to make machine learning fair and ethical

For those working in data science and artificial intelligence with algorithms, you'll find a few ways to make sure that machine learning is fair and ethical. You can:

- Examine the algorithms' ability to influence human behaviour and decide whether it appears biassed. Then, create algorithmic methods that avoid predictive bias.

- Identify any vulnerabilities or inconsistencies in public data sets, and assess whether there is a privacy violation.

- Utilise tools that can help prevent and eliminate bias in machine learning.

Fairness in Machine Learning

Tools for machine learning fairness

Plenty of courses, tools, and processes are available to help you integrate machine learning fairness into your organisation's workflow and prevent machine learning malpractice. A few of these tools include:

IBM's AI Fairness 360: This is a Python toolkit of technical solutions on fairness metrics and algorithms that helps users and researchers share and evaluate discrimination and bias in machine learning models.

Google's What-If Tool: This visualisation tool explores a model's performance on a data set, assessing against preset definitions of fairness constraints. It supports binary classification, multi-class classification, and regression tasks.

Google's Model Cards and Toolkit: This tool confirms that a given model's intent matches its use case and helps users understand the conditions under which their model is safe and appropriate to proceed with.

Microsoft's fairlearn.py: This is an open-source Python toolkit that assesses and improves fairness in machine learning. With an interactive visualisation dashboard and unfairness mitigation algorithms, this tool helps users analyse the trade-offs between fairness and model performance.

Deon: This ethics checklist facilitates responsible data science by evaluating and systematically reviewing applications for potential ethical implications, from the early stages of data collection to implementation.